

HW 1

1

a

Let a and b be Boolean variables. Then $a \cdot b = 0$ and $a + b = 1$ implies that $a = b'$.

✓ Answer ✓

True

$$a \cdot b = 0 \implies a \text{ or } b = 0$$

$$a + b = 1 \implies a \text{ or } b = 1$$

This means that one of a or b must be 1 and one must be 0.

$$\implies a = b'$$

b

Let X and Y be Boolean expressions. Then $X \cdot Y = 0$ and $X + Y = 1$ implies $X = Y'$.

✓ Answer

True by the same logic, as expressions evaluate to boolean variables ultimately.

2

a

Given $f(w, x, y, z) = w + (x + yz)(y' + xz)$, find a sum-of-products expression for f by "multiplying out" this expression.

✓ Answer

$$f(w, x, y, z) = w + xy' + xxz + yy'z + yzxx$$

$$= w + xy' + xz + xyz$$

$$= w + xy' + xz$$

b

Write the dual expression of $\text{Dual}[f(w, x, y, z)]$ given in part (a).

✓ Answer

$$w((x(y + z)) + (y'(x + z)))$$

c

Find a product-of-sums expression for f by multiplying out the expression of part (b) and then taking the dual.

✓ Answer

$$\begin{aligned} & wxy + wxz + wy'x + wy'z \\ &= wx + wxz + wy'z \\ &\rightarrow (w + x)(w + x + z)(w + y' + z) \end{aligned}$$

3

An indicator circuit is to be designed for a room that has two swinging doors $D1$, and $D2$. Associated with each door there are two switches $e1$ and $x1$ for $D1$, $e2$ and $x2$ for $D2$. The ei switch is closed only when the corresponding Di is open in, and the xi switch is closed only when the corresponding Di is open out. As indicator lamp is to be lit whenever there is a clear path through the room (one door opens in and the other door opens out). Fill out a table of combinations for the function corresponding to the indicator light.

✓ Answer

Since for a corresponding ei and xi , only one can be on at any time, we will treat them as part of the same state.

Indicator	$e1'x1'$	$e1$	$x1$
$e2'x2'$	0	0	0
$e2$	0	0	1
$x2$	0	1	0

4

Write the canonical sum and the canonical product for

a

$$f(x, y, z) = \sum(0, 3, 6)$$

✓ Answer

$$\begin{aligned} f &= x'y'z' + x'yz + xyz' \\ &= \prod(1, 2, 4, 5, 7) \\ f &= (x + y + z')(x + y' + z)(x' + y + z)(x' + y + z')(x' + y' + z') \end{aligned}$$

b

$$f(x, y, z) = \prod(1, 2, 7)$$

✓ Answer

$$f = (x + y + z')(x + y' + z)(x' + y' + z')$$

$$= \sum(0, 3, 4, 5, 6)$$

$$f = x'y'z' + x'yz + xy'z' + xy'z + xyz'$$

5

a

If $f = x \oplus y$, express f in terms of $x, x', y, y', +, \cdot$.

✓ Answer

$$x'y + xy'$$

b

Prove that $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

✓ Answer

$$\begin{aligned} &= (x'y + xy')'z + (x'y + xy')z' \\ &= ((x + y')(x' + y))z + x'yz' + xy'z' \\ &= xyz + x'y'z + x'yz' + xy'z' \\ &= x(yz + y'z') + x'(y'z + yz') \\ &= x(y'z + yz')' + x'(y'z + yz') \\ &= x(y \oplus z)' + x'(y \oplus z) \\ &= x \oplus (y \oplus z) \end{aligned}$$

c

Prove that if $x \oplus y = z$, then $x \oplus z = y$ and $x \oplus y \oplus z = 0$.

✓ Answer

$$\begin{aligned} &x \oplus z \\ &= x \oplus x \oplus y \\ &= 0 \oplus y \\ &= y \end{aligned}$$

$$\begin{aligned} &x \oplus y \oplus z \\ &= z \oplus z \end{aligned}$$

$$= zz' + z'z$$

$$= 0$$

d

Prove that $x \oplus y = x + y$ if $xy = 0$

✓ **Answer**

$$x \oplus y$$

$$x'y + xy'$$

$$= (xy + x'y')'$$

$$= (0 + x'y')'$$

$$= (x'y')'$$

$$= x + y$$

e

Prove that $(a + b) \oplus (a + c) = a'(b \oplus c)$.

✓ **Answer**

$$(a + b)(a + c)' + (a + b)'(a + c)$$

$$(a + b)a'c' + a'b'(a + c)$$

$$aa'c' + ba'c' + aa'b' + ca'b'$$

$$ba'c' + ca'b'$$

$$a'(bc' + cb')$$

$$a'(b \oplus c)$$

f

Prove that $x + y = x \oplus y \oplus xy = x \oplus x'y$

✓ **Answer**

$$x \oplus y \oplus xy$$

$$(xy' + x'y) \oplus xy$$

$$(xy' + x'y)(xy)' + (xy' + x'y)'(xy)$$

$$(xy' + x'y)(x' + y') + (xy + x'y')(xy)$$

$$(xy' + x'y) + xy$$

$$x(y + y') + y + (x + x')$$

$$x + y$$

$$x \oplus x'y$$

$$x(x'y)' + x'(x'y)$$

$$x + xy' + x'y$$

$$x + x'y$$

$$x + x(y + y') + x'y$$

$$x + xy + xy' + x'y$$

$$x + y(x + x')$$

$$x + y$$

6

Prove the following identities without using perfect induction.

a

$$ab' + bc' + ca' = a'b + b'c + c'a$$

✓ **Answer**

$$ab' + bc' + ca'$$

$$ab'(c + c') + (a + a')bc' + a'(b + b')c$$

$$ab'c + ab'c' + abc' + a'bc' + a'bc + a'b'c$$

$$(a'bc' + a'bc) + (ab'c + a'b'c) + (ab'c' + abc')$$

$$a'b(c' + c) + b'c(a + a') + c'a(b' + b)$$

$$a'b + b'c + c'a$$

b

$$ab + a'c + bcd = ab + a'c$$

✓ **Answer**

$$ab + a'c$$

$$b = c'$$

$$ab + a'b'$$

$$ab + a'b' + bb'd$$

$$ab + a'c + bcd$$

7

Find the complements f' of the following functions.

Prove your answers are correct by showing that $ff' = 0$ and $f + f' = 1$

a

$$f = a + bc$$

✓ **Answer**

$$a'(b' + c')$$

$$\begin{aligned}
 & a'(b' + c')(a + bc) \\
 & (a'b' + a'c')(a + bc) \\
 & aa'b' + aa'c' + a'b'bc + a'c'bc \\
 & = 0
 \end{aligned}$$

$$\begin{aligned}
 & a'(b' + c') + a + bc \\
 & a'b' + a'c' + a + bc \\
 & a'b' + a'c' + a + abc + a'bc \\
 & a'(b' + c' + bc) + a \\
 & a'((bc)' + bc) + a \\
 & a' + a \\
 & = 1
 \end{aligned}$$

b

$$f = (a + b)(a'c + d)$$

✓ **Answer**

$$\begin{aligned}
 & a'b' + d'(a + c') \\
 & (a'b' + d'(a + c'))(a + b)(a'c + d) \\
 & (a'b' + d'a + d'c')(aa'c + ad + ba'c + bd) \\
 & (a'b' + d'a + d'c')(ad + ba'c + bd) \\
 & a'b'ad + d'aad + d'c'ad + a'b'ba'c + d'aba'c + d'c'ba'c + a'b'bd + d'abd + d'c'bd \\
 & = 0
 \end{aligned}$$

$$\begin{aligned}
 & a'b' + d'(a + c') + (a + b)(a'c + d) \\
 & a(d' + d'c' + d + bd) + a'(b' + d'c' + bc + bd) \\
 & a(d' + d) + a'(b' + b(d'c' + c + d)) \\
 & a + a'(b' + b((c + d)' + (c + d))) \\
 & a + a'(b' + b) \\
 & a + a' \\
 & = 1
 \end{aligned}$$

c

$$f = ab + b'c + ca'd$$

✓ **Answer**

$$\begin{aligned}
 & (a' + b')(b + c')(c' + a + d') \\
 & (a' + b')(b + c')(c' + a + d')(ab + b'c + ca'd) \\
 & (a' + b')(c' + ba + bd')(ab + b'c + ca'd) \\
 & (c'a' + c'b' + a'bd')(ab + b'c + ca'd) \\
 & c'a'ab + c'b'ab + a'bd'ab + c'a'b'c + c'b'b'c + a'bd'b'c + c'a'ca'd + c'b'ca'd + a'bd'ca'd \\
 & = 0
 \end{aligned}$$

$$\begin{aligned}
& (a' + b')(b + c')(c' + a + d') + ab + b'c + ca'd \\
& (a' + b')(c' + ba + bd') + ab + b'c + ca'd \\
& c'a' + c'b' + a'bd' + ab + b'c + ca'd \\
& a(b + b'c + c'b') + a'(c' + bd' + b'c + cd) \\
& a(b + b'(c + c')) + a'(c' + c(bd' + b' + d)) \\
& a(b + b') + a'(c' + c(bd' + (bd')')) \\
& a + a'(c' + c) \\
& a + a' \\
& = 1
\end{aligned}$$

8

Are the following identities valid. Do not use perfect induction.

a

$$ab + c'd' + a'bcd' + ab'c'd' = (a + d')(b + c')$$

✓ Answer

$$\begin{aligned}
& ab + c'd' + a'bcd' + ab'c'd \\
& ab + c'd' + a'bc'd' + a'bcd' + ab'c'd \\
& ab + c'd' + a'bd' + ab'c'd \\
& ab + c'd' + abd' + a'bd' + ab'c'd \\
& ab + c'd' + bd' + ab'c'd \\
& ab + c'd' + bd' + abc'd + ab'c'd \\
& ab + c'd' + bd' + ac'd \\
& ab + c'd' + bd' + ac'd' + ac'd \\
& ab + c'd' + bd' + ac' \\
& a(b + c') + d'(b + c') \\
& (a + d')(b + c')
\end{aligned}$$

b

$$(a + b')(b + c')(c + a') = (a' + b)(b' + c)(c' + a)$$

✓ Answer

$$\begin{aligned}
& (a + b')(b + c')(c + a') \\
& (a'b + b'c + c'a)' \\
& \text{See part 6a} \\
& (ab' + bc' + ca')' \\
& (a' + b)(b' + c)(c' + a)
\end{aligned}$$

c

$$(a + b)(b + c)(c + a) = (a' + b')(b' + c')(c' + a')$$

✓ Answer

False, see $(a, b, c) = (1, 1, 1)$

$$(1 + 1)(1 + 1)(1 + 1) = 1$$

$$(0 + 0)(0 + 0)(0 + 0) = 0$$

d

$$ab + a'b'c = (c + a)(c + b)$$

✓ Answer

False, see $(a, b, c) = (1, 0, 1)$

$$0 + 0 = 0$$

$$1 * 1 = 1$$

9

Let $f(w, x, y, z) = \sum(5, 7, 15)$ and $g(w, x, y, z) = \prod(1, 3, 4, 6, 9, 11, 12, 14)$ Using a Karnaugh map, determine the functions h such that $f = gh'$. How many completely specified functions h solve the problem?

✓ Answer

Term	f	g	h
0	0	1	1
2	0	1	1
5	1	1	0
7	1	1	0
8	0	1	1
10	0	1	1
13	0	1	1
15	1	1	0

Where the rest has both f and $g = 0$

There are 8 free values here, making h have 8 degrees of freedom, with 2 options per degree.

2^8 possible distinct definitions of h

You are presented with a set of requirements under which an insurance policy will be issued. The applicant must be:

1. A married female 25 years old or over, or
2. A female under 25, or
3. A married male under 25 who has not been involved in a car accident, or
4. A married male who has been involved in a car accident, or
5. A married male 25 years or over who has not been involved in a car accident

The variables w, x, y, z assume the truth value 1 in the following cases:

- $w = 1$ if applicant has been involved in a car accident
- $x = 1$ if applicant is married
- $y = 1$ if applicant is a male
- $z = 1$ if applicant is under 25

a

Find an expression which assumes the value 1 whenever the policy should be issued.

✓ **Answer**

$$xy'z' + y'z + xyzw' + xyw + xyz'w'$$

b

Simplify algebraically the above expression and suggest a simpler set of requirements.

✓ **Answer**

$$xy'z' + y'z + xy(w + w'(z + z'))$$

$$xy'z' + y'z + xy(w + w')$$

$$y'(xz' + z) + xy$$

$$y'(x + z) + xy$$

$$y'x + y'z + xy$$

$$x + y'z$$

Married, or a female under 25.